



AI VISIBILITY RESEARCH WORKING PAPER

A Framework for Evaluating AI Search Visibility Signals

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This paper defines a framework for evaluating AI visibility through four outcome layers: AI referral traffic, AI search inclusion, AI brand mentions, and AI recommendation behaviour. It separates these outcome signals from diagnostic signals such as source page count, entity clarity, authority and trust, and technical readiness, then applies the model to a controlled university prompt dataset across multiple AI systems. The dataset and prompt testing environment used in this paper were developed through applied work within Hunterlodge Advertising.

Keywords

AI visibility, AI search, AI-generated answers, source selection, attribution behaviour, recommendation behaviour, entity clarity, trust signals, generative search

I. INTRODUCTION

Search visibility has historically been measured through rankings, impressions, traffic, and conversions. A page appeared in a search result, a user clicked, and performance was recorded through analytics systems that were built around visits rather than generated answers.

AI-generated answers change that sequence because the system produces a response before the user reaches a website. The system interprets the query, selects information, blends sources, and decides what to include in the answer, which means visibility can be gained or lost before the click exists.

This creates a measurement problem for organisations that still treat search visibility as a ranking or traffic issue. A site may rank well but not be selected, a source may shape an answer without being credited, and a brand may be recommended without receiving immediate referral traffic.

The purpose of this paper is to define a framework that measures those behaviours directly. The framework separates observable AI behaviour from the signals that explain that behaviour, which allows AI visibility to be analysed as a distinct layer from traditional SEO.

II. WHAT STANDARD SEO METRICS FAIL TO EXPLAIN

Traditional SEO metrics remain useful, but they do not explain every visibility event inside AI-generated answers. Ranking position measures placement in a results list, while AI-generated answers depend on source interpretation, source selection, and response construction.

Traffic also misses part of the problem because it only records behaviour after a user visits a site. When an AI answer satisfies the query directly, influence may occur without a visit, and the page that shaped the response may receive no measurable session at all.

This means a brand can lose influence inside an answer while its rankings appear stable. It also means a lower-ranked or secondary source can gain visibility if its content is easier for the system to extract, summarise, and reuse.

The measurement gap is therefore not caused by SEO becoming irrelevant. It is caused by measuring a retrieval-era output while users increasingly encounter generated answers that collapse retrieval, interpretation, and recommendation into one response.

III. THE AI VISIBILITY MODEL

The framework defines four observable layers of AI visibility. Each layer describes a different type of behaviour, so the layers should not be merged into a single undifferentiated score.

A. AI Referral Traffic

AI referral traffic measures visits that originate from AI systems. It is useful because it shows that an answer created enough interest for a user to leave the AI environment and visit a site.

The limitation is that referral traffic appears at the end of the interaction. A source can influence an answer without receiving the visit, so traffic should be treated as one visibility outcome rather than the main measure of AI visibility.

B. AI Search Inclusion

AI search inclusion measures whether a page, domain, or source appears inside an AI-generated response. It captures selection behaviour, which is the point where a source becomes part of the answer rather than remaining only part of the wider index.

Inclusion matters because a page that is not selected cannot influence the generated answer. This remains true even when that page ranks in traditional search results for similar queries.

C. AI Brand Mentions

AI brand mentions measure whether a brand or entity is named in the generated response. Mentions capture attribution behaviour, where the system connects information to a visible entity rather than only using information in the background.

This distinction matters because inclusion and attribution are not the same behaviour. A system may use information from a source while failing to name the organisation if entity clarity is weak or if the answer blends multiple sources into a general summary.

D. AI Recommendation Behaviour

AI recommendation behaviour measures whether a brand is actively suggested when the user asks for options, comparisons, or decisions. This is the strongest behavioural signal because the system is not only retrieving information but helping shape a choice.

Recommendation behaviour depends on more than relevance. The system needs enough confidence in the entity, topic association, and trust signals before it presents one source or brand as a credible option.

IV. OUTCOME SIGNALS AND DIAGNOSTIC SIGNALS

The framework separates outcome signals from diagnostic signals. Outcome signals describe what the AI system does, while diagnostic signals explain why that behaviour is likely to occur.

Outcome signals include inclusion, mentions, recommendations, and referral traffic. These signals are observed directly from AI outputs and analytics data, so they describe visible behaviour rather than assumed causes.

Diagnostic signals include source page count, entity clarity, authority and trust, and technical readiness. These signals influence whether a system can identify the source, understand the entity, validate the claim, and reuse the information safely inside a generated answer.

This separation makes the model usable because the same outcome can have different causes. A university may be absent because the system cannot access the content, because the entity is unclear, because other sources carry stronger authority, or because the content does not match the query format well enough to be reused.

Layer What it observes Primary interpretation

Referral traffic Visits from AI systems post-answer user action Search inclusion Source appears in response Selection behaviour

Brand mentions Entity is named Attribution behaviour Recommendation Entity is suggested Decision influence

Layer	What is observed	Interpretation
Referral traffic	Visits from AI systems	Post-answer user action
Search inclusion	Source appears in response	Selection behaviour
Brand mention	Entity is named	Attribution behaviour
Recommendation	Entity is suggested	Decision influence

Table 1. Observable AI visibility layers and their interpretation.

V. HOW THE SIGNALS BEHAVE IN PRACTICE

AI search inclusion depends on whether the system can interpret and reuse the content without excessive reconstruction. These behaviours do not operate independently, which is why separating outcome signals from diagnostic signals is necessary to understand how AI visibility is formed.

Clear headings, direct answers, consistent terminology, and topic-specific pages increase the chance that a source can be selected.

AI brand mentions depend on whether the system can connect information to a stable entity. If the same institution, brand, or author is described differently across pages and platforms, the system has less confidence when assigning attribution.

AI recommendation behaviour depends on repeated reinforcement across sources. A university may be included because it explains a topic clearly, but it is more likely to be recommended when multiple sources reinforce the same association between the university, the subject, and the user need.

AI referral traffic depends on whether the generated answer leaves the user with a reason to click. If the answer fully resolves the query, referral traffic may remain low even where inclusion is high, which is why traffic cannot be treated as the only sign of AI visibility.

VI. MEASUREMENT METHOD

Measurement begins by grouping prompts by intent. Discovery prompts test whether a system can find and surface relevant sources, comparison prompts test how the system separates options, and decision prompts test which sources are recommended when the user is closer to choosing.

Each prompt should preserve a clear user intent while allowing controlled variation in wording. This matters because AI systems can change their source selection when the phrasing

changes, even when the underlying need remains similar.

For every output, the researcher records the primary brand or source, secondary brands or sources, the answer type, the source type, and the observable behaviour. This creates a structured record that can be compared across systems rather than a loose collection of screenshots.

Results are then compared across systems, query types, and repeated source appearances. The goal is to identify stable patterns, such as repeated inclusion across discovery prompts or repeated recommendation in decision prompts, rather than treating one response as proof of visibility.

The prompt dataset and structured observation model were developed through ongoing AI visibility analysis conducted inside Hunterlodge Advertising.

VII. EXAMPLE: UNIVERSITY PROMPT TESTING

This section applies the framework to a controlled university dataset to demonstrate how AI visibility behaves in a real-world environment.

A Computer Science prompt set was tested across ChatGPT, Google AI Overview, Perplexity, Claude, Gemini, and Grok. The prompts were designed around course structure, programming experience, workload, teaching style, skills, career outcomes, and decision support.

The dataset recorded the AI system, intent, prompt, primary brand, secondary brands, source type, answer summary, and observation. This structure makes it possible to compare how different systems select universities for similar student questions.

A. Prompt Set Used

- What universities explain Computer Science course structure clearly?
- Which universities explain how much programming experience is expected?
- Which universities explain workload expectations in Computer Science?

- Which universities explain career outcomes for Computer Science students?
- Which universities explain first year Computer Science clearly?
- Which universities help students compare Computer Science courses?
- Which universities explain differences in Computer Science teaching styles?
- Which sources help students choose between Computer Science options?
- Which universities explain the skills gained from a Computer Science degree?
- Which universities compare Computer Science course content clearly?
- Which universities explain how Computer Science is taught?
- Which universities help students compare Computer Science options?
- Which universities explain different Computer Science teaching methods?
- Which universities explain workload and pacing in Computer Science?
- Which universities help students compare Computer Science course options before choosing?
- Which universities help students choose Computer Science courses?
- Which universities explain how Computer Science works in practice?
- Which universities help students compare Computer Science teaching styles?

B. Observed Pattern

The results show that visibility concentrates around a small group of universities. When primary and secondary appearances are counted once per prompt result, University of Bath and University of York each appear in 14 of 21 outputs, University of Cambridge appears in 13, University of Oxford appears in 11, and University of St Andrews appears in 11.

Primary-source selection shows a different layer of behaviour. University of Cambridge and University of York each appear as the primary brand in four outputs, while University of Bath,

University of Sussex, and University of St Andrews each appear as primary brands in three outputs.

The difference between total appearances and primary appearances is important. A university may be part of the answer set without being positioned as the leading source, which shows why inclusion, mention, and recommendation behaviour need to be recorded separately.

Source	Total appearances	Primary appearances	Pattern
University of Bath	14/21	3/21	High inclusion, weaker primary selection
University of York	14/21	4/21	High inclusion, strong decision
University of Cambridge	13/21	4/21	High inclusion, frequent primary selection
University of Oxford	11/21	2/21	Consistent inclusion, lower primary selection
University of St Andrews	11/21	3/21	Consistent inclusion across systems

Table 2. Summary of repeated university visibility in the Computer Science prompt set.

C. Framework Interpretation

The repeated presence of Bath, York, Cambridge, Oxford, and St Andrews shows that AI visibility is not evenly distributed across all possible universities. The same sources recur across systems because their content, entity recognition, and external reinforcement are easier for systems to reuse.

The dataset also shows that decision queries behave differently from discovery queries. When the prompt asks which, sources help a student choose list-style comparison sources such as Complete University Guide appear because they already organise multiple options in a decision-friendly format.

This pattern supports the framework because it separates the visible outcome from the underlying explanation. The outcome is repeated inclusion and primary selection for a

limited set of universities, while the diagnostic explanation points to source structure, entity clarity, authority reinforcement, and answer-ready content.

VIII. LIMITATIONS OF THE MODEL

Results depend heavily on how queries are phrased, even when intent remains similar. Small changes in wording can shift which sources are selected, which affects inclusion, mentions, and recommendation behaviour across runs. Different AI systems also produce different outputs because they rely on different retrieval methods, data access, ranking logic, and generation behaviour. A university may appear consistently in one system and less consistently in another, which means visibility should be assessed across more than one environment.

Outputs change over time as models are updated and new information becomes available. A university that appears consistently in one test period may lose visibility later if

competing sources improve their content structure or external reinforcement.

These limitations do not invalidate the model, but they define how it should be used. The framework is strongest when applied as repeated measurement across prompt sets, systems, and time periods rather than as a single snapshot.

IX. CONCLUSION

AI visibility is not a ranking position, a traffic source, or a single score. It is the result of how AI systems interpret queries, select sources, assign attribution, and generate recommendations. Although demonstrated using universities, the framework applies to any domain where AI-generated answers influence discovery and decision-making.

The framework presented in this paper separates outcome signals from diagnostic signals so that AI visibility can be measured and explained. This distinction matters because a

brand may be included, mentioned, recommended, or clicked for different reasons.

The university prompt example shows how the framework works in practice. Repeated appearances across multiple AI systems reveal source concentration, while differences between total inclusion and primary selection show why AI visibility needs more precise measurement than traditional ranking or traffic reports provide.

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